

Problem

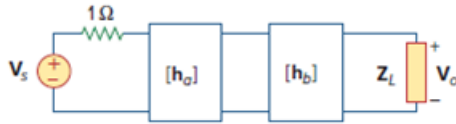
A two-stage amplifier in Fig. 19.134 contains two identical stages with

$$[h] = \begin{bmatrix} 2 \text{ k}\Omega & 0.004 \\ 200 & 500 \mu\text{S} \end{bmatrix}$$

If $Z_L = 20 \text{ k}\Omega$, find the required value of V_s to produce

$V_o = 16 \text{ V}$.

Figure 19.134



Step-by-step solution

Step 1 of 7

Consider the following two-stage amplifier circuit diagram:

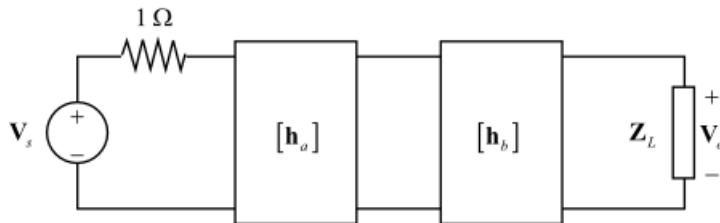


Figure 1

Step 2 of 7

Consider the following data for the circuit in Figure 1.

The following are the h parameters of the identical stages h_a and h_b :

$$[h] = \begin{bmatrix} 2 \text{ k}\Omega & 0.004 \\ 200 & 500 \mu\text{S} \end{bmatrix}$$

The load impedance, $Z_L = 20 \text{ k}\Omega$.

The output voltage, $V_o = 16 \text{ V}$.

Step 3 of 7

Calculate the determinant of h parameters of the identical stages.

$$\begin{aligned}\Delta_h &= h_{11}h_{22} - h_{12}h_{21} \\ &= (2 \text{ k}\Omega)(500 \text{ }\mu\text{S}) - (0.004)(200) \\ &= 1 - 0.8 \\ &= 0.2\end{aligned}$$

The identical stages h_a and h_b are cascaded in the circuit of Figure 1. The T parameters are multiplied in the correct order in the cascade connection.

Convert the h parameters to T parameters.

$$\begin{aligned}[\mathbf{T}] &= \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{\Delta_h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{0.2}{200} & -\frac{2 \text{ k}\Omega}{200} \\ -\frac{500 \text{ }\mu\text{S}}{200} & -\frac{1}{200} \end{bmatrix} \\ &= \begin{bmatrix} -0.001 & -10 \text{ }\Omega \\ -2.5 \text{ }\mu\text{S} & -0.005 \end{bmatrix}\end{aligned}$$

Step 4 of 7

The following are the two T parameters for the two identical stages.

$$\begin{aligned}[\mathbf{T}_a] &= \begin{bmatrix} -0.001 & -10 \text{ }\Omega \\ -2.5 \text{ }\mu\text{S} & -0.005 \end{bmatrix} \\ [\mathbf{T}_b] &= \begin{bmatrix} -0.001 & -10 \text{ }\Omega \\ -2.5 \text{ }\mu\text{S} & -0.005 \end{bmatrix}\end{aligned}$$

Multiply the T parameters of two identical stages to get a single T parameter for the cascaded stages.

$$\begin{aligned}[\mathbf{T}] &= [\mathbf{T}_a][\mathbf{T}_b] \\ &= \begin{bmatrix} -0.001 & -10 \text{ }\Omega \\ -2.5 \text{ }\mu\text{S} & -0.005 \end{bmatrix} \begin{bmatrix} -0.001 & -10 \text{ }\Omega \\ -2.5 \text{ }\mu\text{S} & -0.005 \end{bmatrix} \\ &= \begin{bmatrix} (1 \times 10^{-6}) + (25 \times 10^{-6}) & 0.01 + 0.05 \\ (2.5 \times 10^{-9}) + (12.5 \times 10^{-9}) & (25 \times 10^{-6}) + (25 \times 10^{-6}) \end{bmatrix} \\ &= \begin{bmatrix} 26 \times 10^{-6} & 0.06 \\ 15 \times 10^{-9} & 50 \times 10^{-6} \end{bmatrix}\end{aligned}$$

Step 5 of 7

Calculate the determinant of T parameters of the identical stages.

$$\begin{aligned}\Delta_T &= \mathbf{AD} - \mathbf{BC} \\ &= (26 \times 10^{-6})(50 \times 10^{-6}) - (0.06)(15 \times 10^{-9}) \\ &= (1.3 \times 10^{-9}) - (0.9 \times 10^{-9}) \\ &= 0.4 \times 10^{-9}\end{aligned}$$

Convert the T parameters to z parameters.

$$\begin{aligned}[\mathbf{z}] &= \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\mathbf{A}}{\mathbf{C}} & \frac{\Delta_T}{\mathbf{C}} \\ \frac{1}{\mathbf{C}} & \frac{\mathbf{D}}{\mathbf{C}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{26 \times 10^{-6}}{15 \times 10^{-9}} & \frac{0.4 \times 10^{-9}}{15 \times 10^{-9}} \\ \frac{1}{15 \times 10^{-9}} & \frac{50 \times 10^{-6}}{15 \times 10^{-9}} \end{bmatrix} \\ &= \begin{bmatrix} 1.734 \times 10^3 & 0.0267 \\ 66.67 \times 10^6 & 3.34 \times 10^3 \end{bmatrix}\end{aligned}$$

Replace the two identical stages with the z parameters and redraw the circuit as follows:

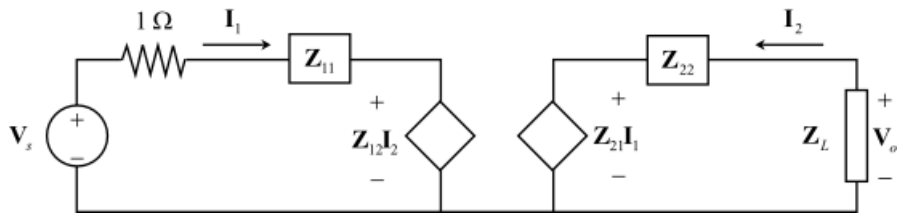


Figure 2

Apply Kirchhoff's voltage law to the first loop of circuit in Figure 2.

$$V_s = (1 \Omega) I_1 + Z_{11} I_1 + Z_{12} I_2$$

$$V_s = (1 \Omega + Z_{11}) I_1 + Z_{12} I_2 \dots\dots (1)$$

Apply Kirchhoff's voltage law to the second loop of circuit in Figure 2.

$$V_o = Z_{22} I_2 + Z_{21} I_1 \dots\dots (2)$$

The following is the output voltage V_o from the circuit in Figure 2.

$$V_o = -I_2 Z_L$$

Therefore,

$$I_2 = \frac{-V_o}{Z_L} \dots\dots (3)$$

Substitute the expression for I_2 from equation (3) in equation (2).

$$\begin{aligned} V_o &= Z_{22} I_2 + Z_{21} I_1 \\ &= Z_{22} \left(\frac{-V_o}{Z_L} \right) + Z_{21} I_1 \\ &= -V_o \left(\frac{Z_{22}}{Z_L} \right) + Z_{21} I_1 \end{aligned}$$

Simplify further to get an expression for the current I_1 .

$$\begin{aligned} Z_{21} I_1 &= V_o + \left(\frac{Z_{22}}{Z_L} \right) V_o \\ I_1 &= \frac{V_o}{Z_{21}} + \left(\frac{Z_{22}}{Z_L} \right) \frac{V_o}{Z_{21}} \\ &= V_o \left(\frac{1}{Z_{21}} + \frac{Z_{22}}{Z_{21} Z_L} \right) \\ I_1 &= V_o \left(\frac{Z_L + Z_{22}}{Z_{21} Z_L} \right) \dots\dots (4) \end{aligned}$$

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Substitute the expressions for I_1 and I_2 from equations (4) and (3) in equation (1).

$$\begin{aligned} V_s &= (1 \Omega + Z_{11}) I_1 + Z_{12} I_2 \\ &= (1 \Omega + Z_{11}) \left(\frac{V_o}{Z_{21}} + \left(\frac{Z_{22}}{Z_L} \right) \frac{V_o}{Z_{21}} \right) + (Z_{12}) \left(\frac{-V_o}{Z_L} \right) \\ &= V_o \left[\frac{(1 \Omega + Z_{11})(Z_L + Z_{22})}{Z_{21} Z_L} - \frac{Z_{12}}{Z_L} \right] \end{aligned}$$

Substitute $20 \text{ k}\Omega$ for Z_L , 16 V for V_o , and the values of z parameters.

$$\begin{aligned} V_s &= (16 \text{ V}) \left[\frac{\left[1 + (1.734 \times 10^3) \right] \left[(20 \times 10^3) + (3.34 \times 10^3) \right]}{(66.67 \times 10^6)(20 \times 10^3)} - \frac{0.0267}{20 \times 10^3} \right] \\ &= (16 \text{ V}) \left[(30.362 \times 10^{-6}) - (1.34 \times 10^{-6}) \right] \\ &= (16 \text{ V}) (29 \times 10^{-6}) \\ &= 464.35 \times 10^{-6} \text{ V} \\ V_s &= 464.35 \mu\text{V} \end{aligned}$$

Thus, the required value of input voltage, V_s is $\boxed{464.35 \mu\text{V}}$.